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CMPUT 365: Introduction to Reinforcement Learning

Worksheet 9: On-policy Prediction with Approximation

Question 1. Let $f(x, y) = (x + y)^2 + e^{xy}$. Recall that the gradient, assuming it exists, is composed of the partial derivatives for each variable,

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f(x,y)}{\partial x} \\ \frac{\partial f(x,y)}{\partial y} \end{bmatrix},$$

where $\frac{\partial f(x,y)}{\partial x}$ is the partial derivative of $f(x, y)$ w.r.t. x assuming that y is fixed.

(a) What is $\nabla f(x, y)$ for the f defined above? Hint: Recall that the derivative of e^z is e^z.

(b) What is $\nabla f(x, y)|_{x=0,y=1}$?

(a) We have

$$\nabla f(x, y) = \begin{bmatrix} \frac{\partial f(x,y)}{\partial x} \\ \frac{\partial f(x,y)}{\partial y} \end{bmatrix} = \begin{bmatrix} 2(x + y) + ye^{xy} \\ 2(x + y) + xe^{xy} \end{bmatrix}.$$

(b) $\nabla f(x, y)|_{x=0,y=1} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Question 2. Find the gradient of f for the following functions:

(a) $f(x, y, z) = \frac{y^z}{x}$, where $x, y, z \in \mathbb{R}, x \neq 0, y > 0$;

(b) $f(x) = e^{x^2+5}$, where $x \in \mathbb{R}$;

(c) $f(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}$, where $\mathbf{x}, \boldsymbol{\beta} \in \mathbb{R}^N$;

(d) $f(\mathbf{x}) = (\mathbf{x}^\top \boldsymbol{\beta} - y)^2$, where $\mathbf{x}, \boldsymbol{\beta} \in \mathbb{R}^N, y \in \mathbb{R}$.

(a)

$$\nabla f(x, y, z) = \begin{bmatrix} -y^z/x^2 \\ zy^{z-1}/x \\ y^z \ln(y)/x \end{bmatrix}.$$

(b)

$$\nabla f(x) = \begin{bmatrix} 2xe^{x^2+5} \end{bmatrix}.$$

(c)

$$\nabla f(\mathbf{x}) = \boldsymbol{\beta}.$$

(d)

$$\nabla f(\mathbf{x}) = 2(\mathbf{x}^\top \boldsymbol{\beta} - y)\boldsymbol{\beta}.$$

Question 3. Exercise 9.1 from Sutton & Barto (2018) (slightly modified). Show that tabular methods (such as those presented in Course 2 of the MOOC and Part I of the RL book) are a special case of linear function approximation. In this case, what would the feature vectors be? Consider for example tabular value function prediction.

Dr. Sutton has asked that the solutions to the textbook exercises not be made electronically available to students. You are encouraged to attend office hours, where the teaching assistants can discuss the exercises with you and show you the answers. Alternatively, if you email Dr. Sutton your answers for an entire chapter, he will likely send you the electronic answer sheet for that chapter.

Question 4. Consider an episodic MDP with the state set $\mathcal{S} = \{1, 2, 3\}$ and a discount rate $\gamma = 0.9$. Suppose $\hat{v}(s; w) = w \cdot s$, where $w \in \mathbb{R}$ and $s \in \mathcal{S}$. The agent observes the following episodic data before reaching the terminal state at time $t = 3$, while following some fixed policy, where $\perp$ denotes the terminal state:

$$S_0 = 1, R_1 = -7, S_1 = 3, R_2 = 5, S_2 = 1, R_3 = 10, S_3 = \perp.$$

Assume that w is initialized to 0 at the beginning of learning and the step-size is fixed to $\alpha = 0.1$. Then

(a) What is $\hat{v}(1; w)$ when w is updated using MOnte-Carlo on this data?

(b) What is $\hat{v}(1; w)$ when w is updated using semi-gradient linear TD on this data?

(c) Again, compute the estimate $\hat{v}(1; w)$ for both gradient Monte-Carlo and semi-gradient Linear TD, when w is updated after seeing the same episode using the respective algorithms.

(Write down the numeric result to two decimal places.)

All calculations are kept to two-decimal precision.

(a) Recall that the Monte-Carlo gradient update is given by

$$w_{t+1} = w_t + \alpha[G - \hat{v}(S_t; w_t)]\nabla_w \hat{v}(S; w_t),$$

where S_t is the state at timestep t and G_t is the corresponding discounted return from that state. Note that, for this question, $\hat{v}(S; w) = wS$ and $\nabla_w \hat{v}(S; w^{(\text{old}} = S$. Let us first compute the returns for each of the states in the given episode:

$$\begin{aligned} G_2 &= R_3 = 10, \\ G_1 &= R_2 + \gamma G_2 = 14, \\ G_0 &= R_1 + \gamma G_1 = 5.6. \end{aligned}$$

Then, the parameter w is updated per time step as follows:

$$\begin{aligned} w_1 &= w_0 + \alpha[G_0 - w_0 S_0]S_0 = 0.00 + 0.1 \cdot [5.6 - 0.00 \cdot 1] \cdot 1 = 0.56, \\ w_2 &= w_1 + \alpha[G_1 - w_1 S_1]S_1 = 0.56 + 0.1 \cdot [14 - 0.56 \cdot 3] \cdot 3 \approx 4.26, \\ w_3 &= w_2 + \alpha[G_2 - w_2 S_2]S_2 = 4.26 + 0.1 \cdot [10 - 4.26 \cdot 1] \cdot 1 \approx 4.83. \end{aligned}$$

Thus, $\hat{v}(1, w_3) = w_3 \cdot 1 = 4.83$.

(b) Recall that the semi-gradient TD update (for state S_t, reward R_t, and next state S_{t+1}) is given by

$$w_{t+1} = w_t + \alpha[R_{t+1} + \gamma \hat{v}(S_{t+1}; w_t) - \hat{v}(S_t; w_t)]\nabla_w \hat{v}(S; w_t).$$

Then, the parameter w' is updated as follows (we use the primed version to represent the weight for the TD case):

$$w'_1 = w'_0 + \alpha[R_1 + \gamma w'_0 S_1 - w'_0 S_0]S_0 = 0.0 + 0.1 \cdot [-7 + 0.9 \cdot 0 \cdot 3 - 0 \cdot 1] \cdot 1 = -0.7,$$
$$w'_2 = w'_1 + \alpha[R_2 + \gamma w'_1 S_2 - w'_1 S_1]S_1 = -0.7 + 0.1 \cdot [5 - 0.9 \cdot 0.7 \cdot 1 + 0.7 \cdot 3] \cdot 3 \approx 1.24,$$
$$w'_3 = w'_2 + \alpha[R_3 + \gamma \hat{v}(\perp; w'_2) - w'_2 S_2]S_2 = 1.24 + 0.1 \cdot [10 + 0.9 \cdot 0 - 1.24 \cdot 1] \cdot 1 \approx 2.12.$$

Thus, $\hat{v}(1; w'_3) = w'_3 \cdot 1 = 2.12$.

(c) We now repeat the above calculations for the two algorithms.

1. Gradient Monte-Carlo:

$$w_4 = 4.83 + 0.1 \cdot [5.6 - 4.83 \cdot 1] \cdot 1 \approx 4.91,$$
$$w_5 = 4.91 + 0.1 \cdot [14 - 4.91 \cdot 3] \cdot 3 \approx 4.69,$$
$$w_6 = 4.69 + 0.1 \cdot [10 - 4.69 \cdot 1] \cdot 1 \approx 5.22.$$

Thus, $\hat{v}(1, w_6) = 5.22$.

2. Semi-gradient TD:

$$w'_4 = 2.12 + 0.1 \cdot [-7 + 0.9 \cdot 2.12 \cdot 3 - 2.12 \cdot 1] \cdot 1 \approx 1.78,$$
$$w'_5 = 1.78 + 0.1 \cdot [5 + 0.9 \cdot 1.78 \cdot 1 - 1.78 \cdot 3] \cdot 3 \approx 2.16,$$
$$w'_6 = 2.16 + 0.1 \cdot [10 + 0.9 \cdot 0 - 2.16 \cdot 1] \cdot 1 \approx 2.94.$$

Thus, $\hat{v}(1, w'_6) = 2.94$.

Question 5. Assume that $\nabla(\mathbf{x})$ exists. Then answer the following using true or false.

(a) The gradient of a function is a function.

(b) The gradient of a function at a point will point in the direction of steepest descent, i.e. following the direction of the gradient will locally decrease the function value.

(c) For a function f, the gradient $\nabla f(\mathbf{x})$ at point $\mathbf{x}$ will be the zero vector if $\mathbf{x}$ globally minimizes f.

(d) For a function f, the gradient $\nabla f(\mathbf{x})$ at point $\mathbf{x}$ will be the zero vector if $\mathbf{x}$ globally maximizes f.

(e) For a function f, if the gradient $\nabla f(\mathbf{x}) = 0$ then $\mathbf{x}$ locally minimizes the function f.

(a) True

(b) False. It follows the direction of steepest ascent.

(c) True

(d) True

(e) False. This condition is necessary but not sufficient, as this particular point can also be a saddle point (think about $f(x) = x^3, \nabla_x f(x) = 3x^2$, which is 0 at $x = 0$). To determine which type of critical point it is, one needs to look at the second derivative (the Hessian matrix $H_f(x)$). It is a local minimum if $H_f(x)$ is positive definite, a local maximum if $H_f(x)$ is negative definite, and it is a saddle point if $H_f(x)$ has mixed signs.